

Introduction to Unsupervised Learning

What is Unsupervised Learning?

Unsupervised learning is a type of machine learning where the model is trained on data without any explicit labels or targets. The goal is to explore the underlying structure of the data, discover patterns, and extract meaningful information.

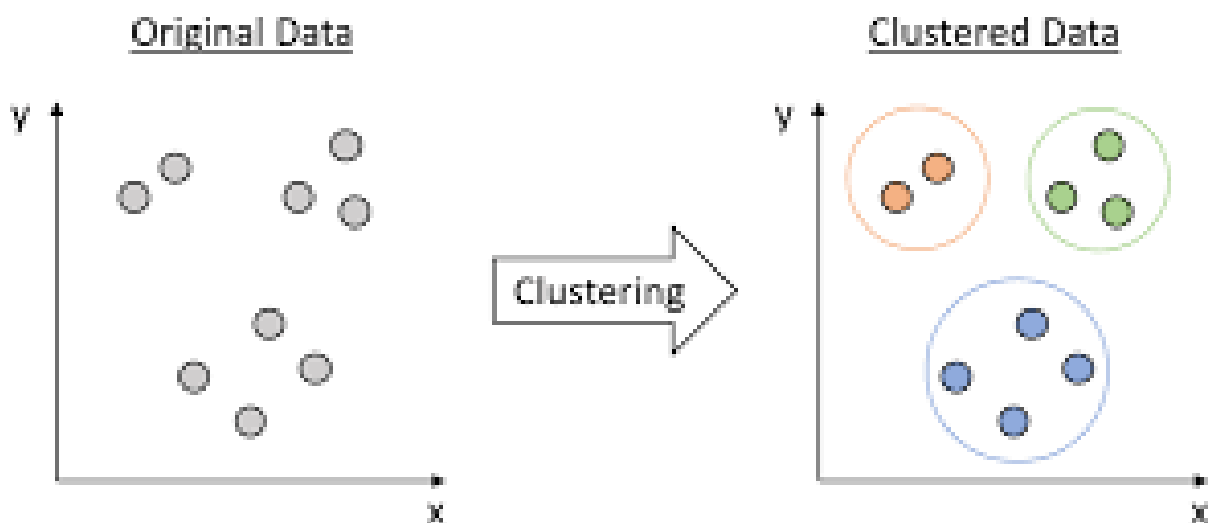
Types of Unsupervised Learning

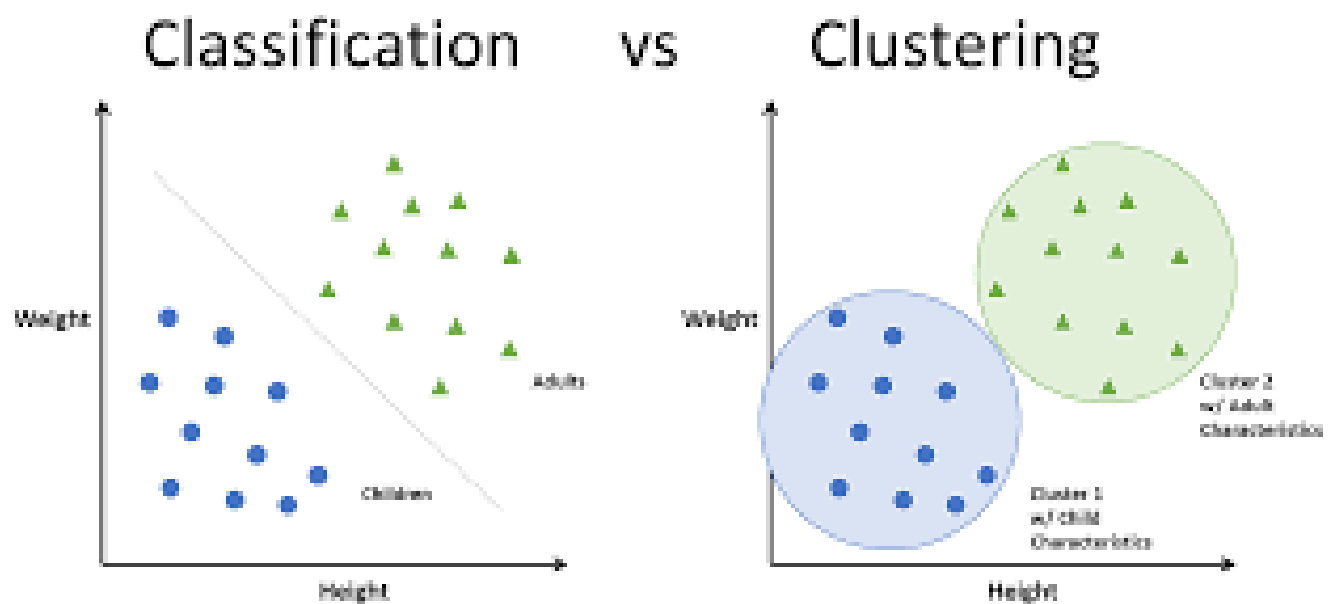
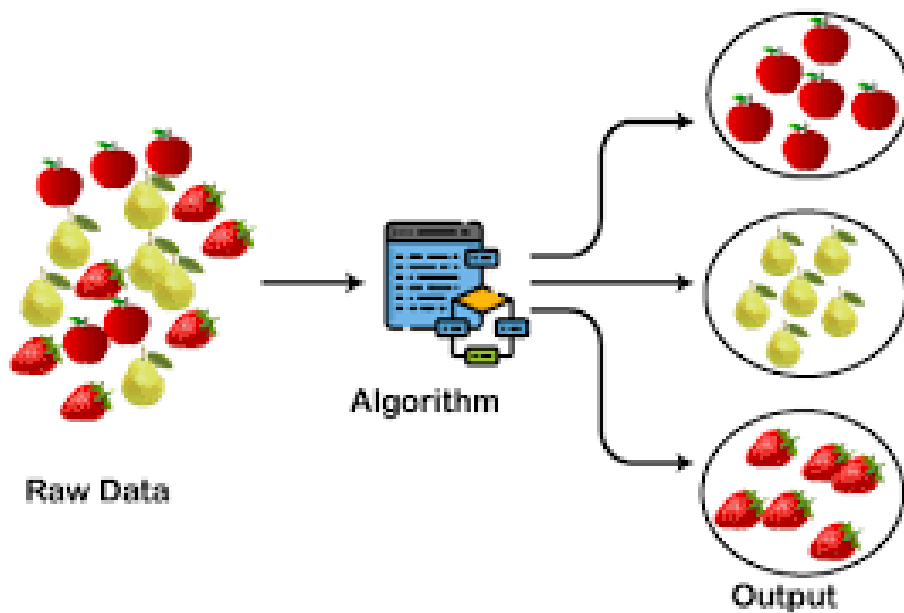
1. **Clustering:** Grouping similar data points into clusters.
2. **Dimensionality Reduction:** Reducing the number of variables while preserving essential information.
3. **Anomaly Detection:** Identifying unusual data points or patterns that differ from the norm.
4. **Association Rule Learning:** Discovering interesting relationships between variables in large datasets.

Clustering

What is Clustering?

Clustering is an unsupervised learning technique used to group similar data points together into clusters based on their features. The goal is to ensure that data points within a cluster are more similar to each other than to those in other clusters.





Choosing the Right Clustering Algorithm

- **K-Means:** Use when clusters are spherical and you know the number of clusters.
- **Hierarchical:** Use for small datasets or when a dendrogram is useful.

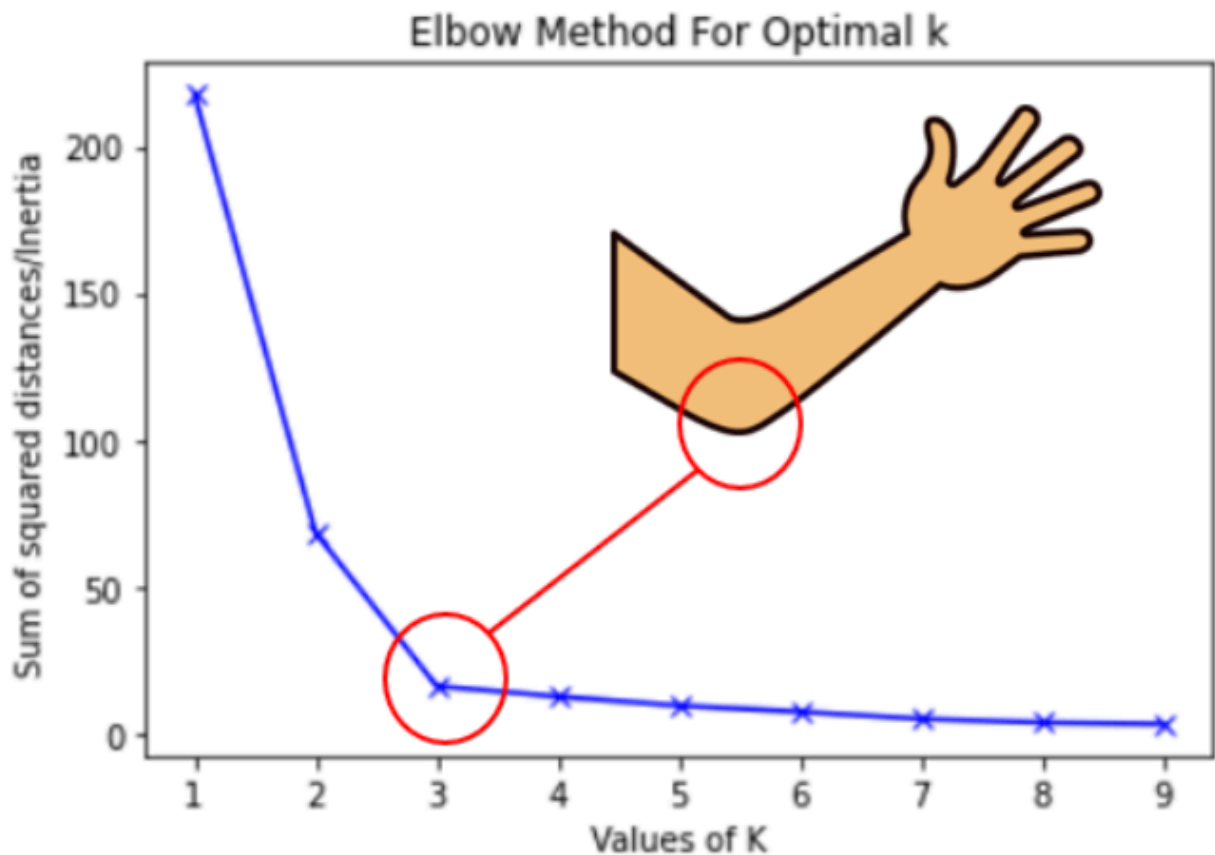
- **DBSCAN:** Use for data with clusters of varying shapes or when detecting noise.
- **GMM:** Use for overlapping clusters or when cluster membership probabilities are needed.

Elbow Method

The elbow method is a technique used to determine the optimal number of clusters for a clustering algorithm, such as K-means. It involves running the clustering algorithm with different numbers of clusters and then plotting the cost (or "inertia") against the number of clusters. The "elbow" point in the plot indicates the optimal number of clusters where adding more clusters does not significantly improve the cost.

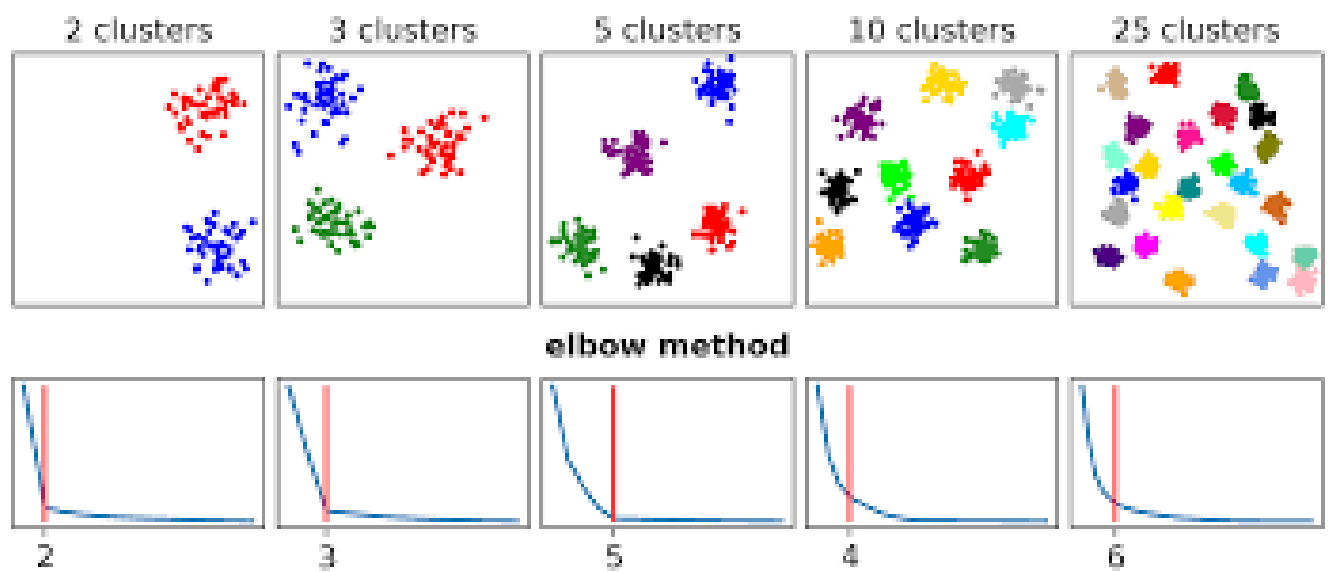
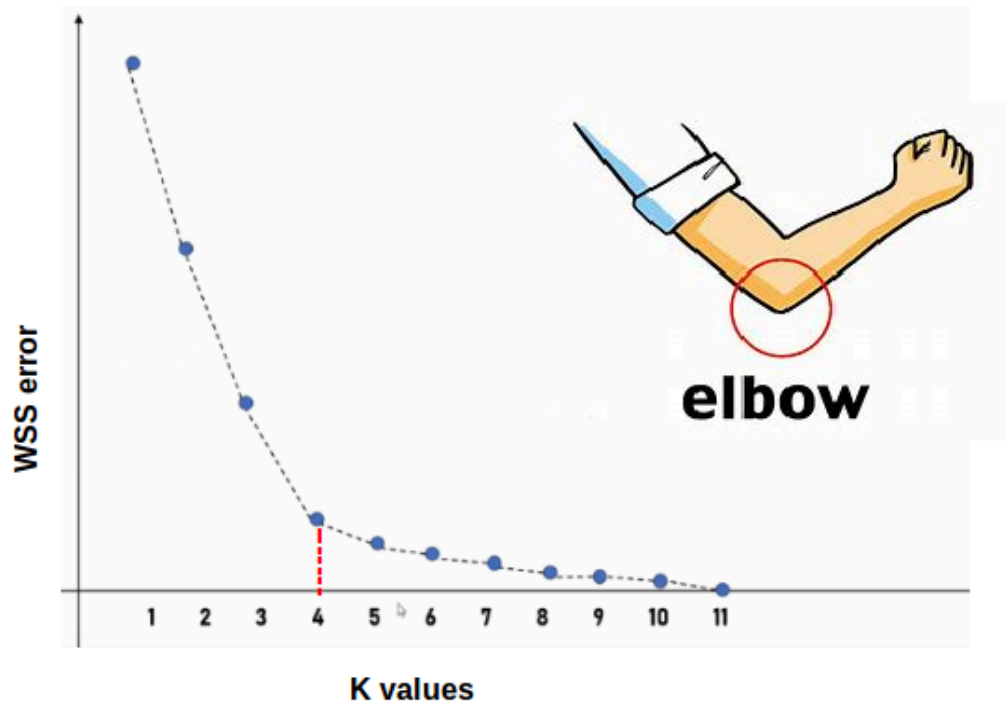
Steps to Implement the Elbow Method

1. **Run K-means Clustering for Different Numbers of Clusters:** Fit the K-means algorithm with a range of cluster numbers.
2. **Calculate the Cost (Inertia):** For each number of clusters, compute the cost (sum of squared distances from each point to its assigned cluster center).
3. **Plot the Cost vs. Number of Clusters:** Create a plot of the cost as a function of the number of clusters.
4. **Identify the Elbow:** Look for the point where the curve bends or levels off, which suggests the optimal number of clusters.



Line plot between K and inertia

Elbow method



The Elbow Method showing the optimal k

